Forecasting VolVue Volatility Proxies with Spillover Effects:
A Zhang-Style GNNHAR Replication and Extension

Volatility Project

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Abstract

This report rewrites the current Dow30, SP100, and SP500 evidence in the style of Zhang, Pu, Cucuringu, and Dong's GNNHAR framework. The central questions are whether graph information improves the heterogeneous autoregressive benchmark, whether nonlinear graph neural aggregation improves on the linear graph HAR model, and whether QLIKE-trained forecasts improve out-of-sample volatility prediction. Our target is not Zhang's high-frequency realized variance. Instead, we use a daily-scaled VolVue 30-day historical-volatility proxy and, in extensions, a daily-scaled 30-day implied-volatility proxy. Under this target, QLIKE training and implied-volatility features are the most stable sources of improvement. The current evidence does not show the same nonlinear GNN gain reported by Zhang et al.; matched DM tests generally do not support GNNHAR1L over GHAR in our completed artifacts, and the larger scale runs show significant deterioration.

1 Introduction

Zhang et al. study realized-volatility forecasting from the perspective of volatility spillovers on a financial graph. Their starting point is the heterogeneous autoregressive model, which captures daily, weekly, and monthly persistence in realized volatility but does not directly use cross-sectional spillover information. They then compare the linear graph HAR model with a customized GNNHAR architecture in which graph neural layers replace the linear one-hop neighborhood aggregation. This gives a natural empirical sequence:

$$\text{HAR} \quad \longrightarrow \quad \text{GHAR} \quad \longrightarrow \quad \text{GNNHAR}.$$

The paper's main empirical emphasis is not only model ranking. It asks whether graph information is useful, whether nonlinear spillovers add predictive power over linear graph aggregation, whether multi-hop neighbors are necessary, and whether QLIKE is a better estimation criterion than MSE for volatility forecasting.

We follow that ordering in this report. The main text therefore focuses on the strict Zhang-core models:

$$\text{HAR}_M,\ \text{GHAR}_M,\ \text{GNNHAR1L}_M,\ \text{GNNHAR2L}_M,\ \text{GNNHAR3L}_M$$

and their QLIKE-trained counterparts where available. Implied-volatility features, four- and five-layer GNNHAR models, and linear multi-hop GHAR variants are reported as extensions. This distinction matters. In Zhang et al., linear GHAR2Hop is an appendix-style diagnostic for the multi-hop question, not the primary competitor to GNNHAR. Accordingly, we do not treat GHAR2H_M or GHAR3H_M as core models in the main comparison.

1.1 Notation

Throughout the report, $i = 1, \ldots, N$ indexes assets and t indexes trading days. The vector $v_t = (RV_{1,t}, \ldots, RV_{N,t})^\top$ denotes the cross-section of volatility targets. In Zhang et al., $RV_{i,t}$ is high-frequency realized variance. In our current data, $RV_{i,t}$ denotes the daily-scaled VolVue historical-volatility proxy unless explicitly stated otherwise.

The graph is $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with binary adjacency matrix A. The normalized spillover matrix is

$$W = D^{-1/2} A D^{-1/2},$$

where D is the degree matrix. The HAR feature matrix $V_{:t-1}$ contains daily, weekly, and monthly lag components. For model names, the subscript M means that the model is trained with MSE as estimation criterion, while Q means that it is trained with QLIKE. The superscript IV denotes the implied-volatility extension.

Loss ratios are written as

$$\rho_M(m) = \frac{L_M(m)}{L_M(\mathrm{HAR}_M)}, \qquad \rho_Q(m) = \frac{L_Q(m)}{L_Q(\mathrm{HAR}_M)}.$$

For older Dow30 and SP500 artifacts whose stored baseline is labeled simply as HAR, the same notation means the ratio relative to that artifact's HAR baseline. The column $1 - \rho_Q$ is the QLIKE gain relative to the relevant HAR baseline; positive values indicate lower QLIKE loss. For pairwise DM tests, A and B denote the first and second models in the comparison, and $1 - L_B/L_A > 0$ favors the second model. The MCS column reports the model confidence set at the 5% level where available.

2 Data and Forecasting Design

Table 1 summarizes the result artifacts used here. The latest completed long-run notebook result is the SP100 A100 run. Dow30 is a completed earlier full-stack result. The SP500 evidence is a short-budget scale artifact rather than a matched A100 long run, so it should be interpreted as preliminary scale evidence.

Universe	N	Dates	Test days	Training budget	Status
Dow30	30	2021-02-26–2026-01-26	186	epochs=250, hidden=16	completed earlier full stack
SP100 scale	99	2021-07-12–2026-06-01	161	epochs=80, hidden=16	short-budget scale artifact
SP500 scale	449	2021-07-19–2026-06-01	184	epochs=60, hidden=16	short-budget scale artifact
SP100 A100	91	2021-06-09–2026-06-09	234	epochs=5000, hidden=9	current Colab long run

Table 1: Available runs. The SP100 A100 run is the latest long-run notebook result; SP500 is not yet a matched A100 long run.

Zhang et al. construct the daily realized variance of stock i on day t from intraday returns:

$$RV_{i,t} = \sum_{\ell=1}^{M} r_{i,t,\ell}^2.$$

Their empirical implementation uses five-minute intraday returns from LOBSTER. Our available VolVue data are different. The current package contains 30-day historical volatility and 30-day mean implied volatility in percentage-point units. To keep the target on a daily variance scale, the notebook applies

$$RV_{i,t}^{\text{proxy}} = \frac{HV30_{i,t}^2}{252}, \qquad IV_{i,t}^{\text{proxy}} = \frac{IV30_{i,t}^2}{252}.$$

This scaling is consistent with a daily percent-squared variance convention, but it is not definition-identical to high-frequency realized variance. The VolVue target is smoother because it is already a rolling 30-day volatility measure. For that reason, differences from Zhang et al.'s reported performance should not be read mechanically as implementation errors. They may reflect a materially different forecasting target.

3 Differences from Zhang et al. and Their Implications

This section is included to prevent a misleading interpretation of the empirical results. We inspected Zhang et al.'s article through the MinerU Markdown extraction and checked the public GitHub implementation cloned under `references/code/GNNHAR`. The source inspection covered volatility construction, HAR/GHAR estimation, GNNHAR training, MCS/DM summaries, regime analysis, and forecast-error plots. The local clone is aligned with the public repository at commit `98a85b0`. The conclusion is that our notebook is Zhang-style in model architecture and evaluation logic, but it is not a definition-identical replication of the published empirical design.

Dimension	Zhang et al. paper and code	Current notebook and report	Effect on interpretation
Volatility target	Daily realized variance from five-minute intraday returns, $RV_{i,t} = \sum_{\ell} r^2_{i,t,\ell}$. The public `compute_vol.py` multiplies five-minute returns by 100, winsorizes each return column at the 0.5% and 99.5% percentiles, and sums squared intraday returns by day.	Daily-scaled VolVue 30-day historical-volatility proxy, $RV^{\text{proxy}}_{i,t} = HV30^2_{i,t}/252$, with $IV30^2_{i,t}/252$ used as an extension feature.	This is the most consequential difference. A rolling $HV30$ proxy is smoother and mechanically more persistent than high-frequency RV. HAR and IV baselines become harder to beat, and nonlinear daily spillover effects can be attenuated. Therefore weaker GNNHAR gains are not by themselves evidence that the implementation is wrong.
Data source and period	LOBSTER intraday data from July 1, 2007 to June 30, 2021, with an out-of-sample period from July 1, 2011 to June 30, 2021. The DJIA sample keeps 27 continuously traded stocks; S&P 100 is used as a robustness universe.	VolVue/Yahoo/Drive panels over the recent five-year window, reported for Dow30, SP100, and SP500-scale universes. The SP100 A100 run is the closest current long run; SP500 remains a short-budget scale artifact.	The market regime, crisis coverage, and graph structure are different. Zhang et al. include the 2008 crisis, 2015–2016 stress, and March 2020; our sample is shorter and concentrated in the post-2021 market. Numerical ratios should not be compared one-for-one across papers.
Forecast horizons	Main tables report one-day, one-week, and one-month horizons. In `compute_vol.py`, horizon targets are formed by summing future daily variance over the horizon.	The current report is mainly a one-step daily-style evaluation of the available VolVue proxy outputs. Complete one-week and one-month Zhang-style tables have not yet been produced.	We cannot yet reproduce Zhang et al.'s horizon conclusion that nonlinear spillovers are strongest at short horizons and weaker at longer horizons. The current evidence only supports claims for the completed daily-style artifacts.
Rolling design	Monthly recalibration with a past-1000-day rolling window. The article describes roughly 36 months for training and the most recent 12 months for validation; the public `GNNHAR.py` also exposes a one-month default validation length through `valid_len=22`, matching the robustness split.	The SP100 A100 run uses the Zhang-style 1000-day lookback, 22-day test window, hidden dimension 9, learning rate 10^{-3}, and one-month validation setup. Some older Dow30 and SP500 artifacts used shorter budgets or legacy settings.	SP100 A100 is the cleanest current comparison. Older Dow30 and SP500 scale artifacts are useful diagnostics, but they should not be treated as fully matched protocol evidence.

Table 2: Data and protocol differences between Zhang et al.'s design and the current VolVue notebook/report.

Dimension	Zhang et al. paper and code	Current notebook and report	Effect on interpretation
Graph construction	`GHAR.py` and `GNNHAR.py` estimate a Graphical Lasso precision matrix on pre-origin daily returns. Edges are the nonzero off-diagonal precision entries, symmetrically degree-normalized as $D^{-1/2}AD^{-1/2}$, with no self-loops.	The notebook and scale pipeline implement the same core GLASSO precision nonzero pattern and symmetric degree normalization. The scale pipeline also contains fallbacks for numerical stability in large panels.	When GLASSO succeeds, the graph-design alignment is strong. If a fallback graph is used, that block should be described as a numerical-stability extension rather than a clean Zhang GLASSO replication.
Model set	Core models are HAR, GHAR, and GNNHAR1L–GNNHAR3L, each under MSE and QLIKE estimation criteria. The S&P 100 robustness section adds four- and five-layer GNNHAR models because the graph diameter can be larger. GHAR2Hop is an appendix diagnostic, not a main benchmark.	We report the strict core separately, then add IV-augmented HAR/GHAR/GNNHAR models, four- and five-layer SP100 extensions, linear multi-hop GHAR diagnostics, and placebo/random-graph checks where available.	IV rows answer a new research question and should not be mixed into the Zhang-core ranking. GHAR2H/GHAR3H are useful for the multi-hop question but should be read as diagnostics, not as replacements for the main HAR–GHAR–GNNHAR sequence.
Hyperparameter tuning	The article sets Adam learning rate 10^{-3}, batch size 32, early stopping, and grid-searches hidden dimension $D^{(\ell)} \in \{3, 6, 9, 16, 32\}$, selecting 9. The public `GNNHAR.py` default uses `batch_size=128`, `n_epochs=5000`, `n_hid=9`, and validation checkpoints.	The latest SP100 A100 run fixes the Zhang-selected hidden dimension 9 and learning rate 10^{-3} with a long training budget. The scale pipeline supports hidden/lr grids, but the current report does not contain a full archived Appendix-D-style tuning table for every universe.	We can say the main SP100 A100 setting follows Zhang's selected hyperparameter point, but we should not claim that every reported artifact has reproduced the full hyperparameter-search appendix. Some GNN underperformance could still be affected by tuning and ensemble choices.
Evaluation outputs	`Summary_Results.py` computes MSE and QLIKE by ticker, normalizes loss ratios, and runs MCS with 10,000 bootstraps. `Summary_Regime.py` splits low/high volatility periods using SPY RV percentiles. `BoxPlot_Error.py` creates forecast-error and forecast-ratio boxplots. The article also reports FVU and DM tests, with QLIKE emphasized for power.	This report includes loss ratios, MCS summaries, matched QLIKE DM tests, SP100 depth DM tests, and linear multi-hop diagnostics. It does not yet include a full SPY-regime section, full forecast-error/ratio figures, MAD over-smoothing figures, or complete multi-horizon FVU tables.	The report is sufficient for a preliminary Zhang-style empirical extension, but it is not yet publication-complete relative to the article's full evidence package. The missing outputs mostly affect explanatory support, not the raw fact that the completed QLIKE DM tests are unfavorable to GNNHAR1L versus GHAR in our current proxy data.

Table 3: Modeling, tuning, and evaluation differences between Zhang et al.'s design and the current VolVue notebook/report.

The strongest implication is about the dependent variable. Zhang et al. forecast high-frequency realized variance, while our current RV^{proxy} is a daily scaling of a 30-day historical-volatility series. Because $HV30$ is itself a rolling statistic, it embeds information from recent returns before the model sees the HAR lags. This makes the baseline persistence structure unusually strong. It also means that option-implied volatility can dominate the strict graph contribution because $IV30$ is a forward-looking market price of volatility risk, whereas Zhang et al.'s baseline experiment deliberately asks what can be learned from realized-volatility and return-based graph information alone.

The second implication is about the meaning of the negative GNNHAR evidence. In Zhang et al., the main empirical statement is that GHAR improves over HAR, GNNHAR1L improves over GHAR at short horizons, deeper GNNs do not monotonically improve forecasts, and QLIKE training improves performance. In our completed artifacts, the QLIKE point transfers most clearly. The nonlinear-spillover point does not: matched QLIKE DM tests are not favorable to GNNHAR1L relative to GHAR, especially once IV is added. Given Tables 2 and 3, the correct interpretation is not that Zhang et al.'s result is contradicted. The correct interpretation is that, for recent VolVue $HV30/IV30$ proxies, QLIKE and implied volatility are more robust sources of predictive improvement than nonlinear one-hop graph aggregation.

The third implication is about how to write the project going forward. A strict replication claim would require reconstructing true daily realized variance from intraday returns, running one-day, one-week, and one-month horizons, preserving the same rolling forecast calendar, archiving the full hidden-dimension tuning record, and reproducing regime, FVU, DM, MCS, forecast-error, forecast-ratio, and over-smoothing outputs. If we keep the current data source, the manuscript should be framed as a Zhang-style extension to 30-day historical- and implied-volatility proxies. That framing is still meaningful, but its economic question is different: whether graph spillovers and neural aggregation add value after a smooth volatility proxy and option-implied information are already available.

4 Methodology

For each asset i, the HAR benchmark uses lagged daily, weekly, and monthly volatility features:

$$x_{i,t}^{(d)} = RV_{i,t-1}, \qquad x_{i,t}^{(w)} = \frac{1}{5}\sum_{\ell=1}^{5} RV_{i,t-\ell}, \qquad x_{i,t}^{(m)} = \frac{1}{22}\sum_{\ell=1}^{22} RV_{i,t-\ell}.$$

The exact averaging convention in the code is the one used by the current notebook pipeline; the role of the three features is the same as in the HAR literature.

The linear graph model augments HAR with neighborhood-aggregated features:

$$\mathbb{E}(v_t \mid \mathcal{F}_{t-1}) = \alpha + V_{:t-1}\beta + WV_{:t-1}\gamma,$$

where $W = D^{-1/2}AD^{-1/2}$ is the symmetrically normalized adjacency matrix without self-loops. Following Zhang et al.'s graph-construction choice, the graph is estimated from pre-origin returns through Graphical Lasso and is then fixed within the corresponding rolling forecast origin.

The GNNHAR model replaces the linear neighborhood term with a nonlinear graph convolution:

$$H^{(\ell+1)} = \mathrm{ReLU}\Big(WH^{(\ell)}\Theta^{(\ell)}\Big), \qquad H^{(0)} = V_{:t-1}.$$

The one-layer model is

$$\mathbb{E}(v_t \mid \mathcal{F}_{t-1}) = \alpha + V_{:t-1}\beta + H^{(1)}\gamma,$$

and the two- and three-layer models replace $H^{(1)}$ by $H^{(2)}$ or $H^{(3)}$. This is the Zhang-style interpretation: one layer tests nonlinear one-hop spillovers; additional layers test whether a larger receptive field adds useful multi-hop information.

The main losses are

$$\mathrm{MSE} = \frac{1}{TN}\sum_{t,i}\big(\widehat{RV}_{i,t} - RV_{i,t}\big)^2$$

and

$$\mathrm{QLIKE} = \frac{1}{TN}\sum_{t,i}\left[\frac{RV_{i,t}}{\widehat{RV}_{i,t}} - \log\left(\frac{RV_{i,t}}{\widehat{RV}_{i,t}}\right) - 1\right].$$

As in Zhang et al., we distinguish the estimation criterion from the forecast loss. HAR_M and $GHAR_M$ are OLS-style MSE-estimated linear models. QLIKE-trained linear models and GNNHAR models are optimized by Adam when the closed-form OLS estimator is unavailable. The latest SP100 A100 run uses the Zhang hyperparameter reference point most relevant to the paper's Appendix D: hidden dimension 9, learning rate 10^{-3}, and the same hidden dimension across deeper layers.

5 Empirical Results

5.1 Dow30

The Dow30 result is useful as a small-universe check of the reporting stack. Table 4 first gives the complete model comparison available in the Dow30 artifact. It includes the strict core, the IV extension, QLIKE-trained extension rows, and diagnostic placebo/random-graph rows. The diagnostic rows are kept in the full table so that the empirical record is complete, but they are not used as Zhang-core evidence. Table 5 then reports the strict core, excluding IV and random-graph diagnostics. The linear graph model improves on HAR under QLIKE, with $\rho_Q = 0.9868$, but the one-, two-, and three-layer GNNHAR rows are worse than HAR in this artifact. This is the opposite of Zhang et al.'s main DJIA finding, where GNNHAR1L improves on GHAR at the daily horizon.

Model	Block	Est.	ρ_M	ρ_Q	$1 - \rho_Q$	ρ_Q^{IV}
GHAR	strict core	MSE	0.9626	0.9868	1.32%	1.0230
HAR	strict core	MSE	1.0000	1.0000	0.00%	1.0368
GNNHAR1L	strict core	MSE	1.0183	1.0304	-3.04%	1.0683
GNNHAR2L	strict core	MSE	1.0039	1.0464	-4.64%	1.0849
GNNHAR3L	strict core	MSE	1.0575	1.0947	-9.47%	1.1350
GNNHAR1L-QLIKE	QLIKE-trained extension	QLIKE	2.3467	1.2320	-23.20%	1.2773
GHAR+IV	IV extension	MSE	0.9114	0.9549	4.51%	0.9901
GNNHAR3L-IV	IV extension	MSE	0.8519	0.9555	4.45%	0.9906
HAR+IV	IV extension	MSE	0.9285	0.9645	3.55%	1.0000
GNNHAR2L-IV	IV extension	MSE	0.8833	0.9993	0.07%	1.0360
GNNHAR1L-IV	IV extension	MSE	0.9199	1.0155	-1.55%	1.0529
GNNHAR1L-IV-QLIKE	IV extension	QLIKE	2.2778	1.1360	-13.60%	1.1778
GHAR+fakeIV	diagnostic placebo	MSE	0.9669	0.9898	1.02%	1.0262
HAR+fakeIV	diagnostic placebo	MSE	0.9953	0.9985	0.15%	1.0352
GNNHAR1L-IV+fakeIV	diagnostic placebo	MSE	1.0457	1.0586	-5.86%	1.0976
GHAR_RANDOM	diagnostic random graph	MSE	0.9672	0.9905	0.95%	1.0269

Table 4: Dow30 full available model comparison. Diagnostic placebo and random-graph rows are included for completeness but are not part of the strict Zhang-core comparison.

Model	ρ_M	ρ_Q	$1 - \rho_Q$
HAR	1.0000	1.0000	0.00%
GHAR	0.9626	0.9868	1.32%
GNNHAR1L	1.0183	1.0304	-3.04%
GNNHAR2L	1.0039	1.0464	-4.64%
GNNHAR3L	1.0575	1.0947	-9.47%
GNNHAR1L$_Q$	2.3467	1.2320	-23.20%

Table 5: Dow30 strict-core loss ratios relative to HAR. The QLIKE-trained row is only available for the legacy one-layer GNN artifact.

Table 6 reports the IV extension for the same universe. Here the strongest rows are GHAR^{IV} and GNNHAR3L^{IV}, with QLIKE gains of 4.51% and 4.45% relative to HAR. The improvement therefore comes mainly from the additional option-implied volatility information and not from a clean one-layer nonlinear spillover effect.

Model	ρ_M	ρ_Q	$1 - \rho_Q$
HAR^{IV}	0.9285	0.9645	3.55%
GHAR^{IV}	0.9114	0.9549	4.51%
GNNHAR1L^{IV}	0.9199	1.0155	-1.55%
GNNHAR2L^{IV}	0.8833	0.9993	0.07%
GNNHAR3L^{IV}	0.8519	0.9555	4.45%
GNNHAR1L$_Q^{IV}$	2.2778	1.1360	-13.60%

Table 6: Dow30 IV-augmented extension loss ratios relative to HAR.

5.2 Current SP100 A100 Run

Table 7 is the cleanest current long-run evidence because it comes from the latest SP100 A100 notebook run. It contains the full available SP100 A100 comparison: strict Zhang-core models, IV extensions, and four-/five-layer depth extensions. Table 8 isolates the strict core. The strict-core results show three facts. First, GHAR$_M$ modestly improves on HAR$_M$, with $\rho_Q = 0.9925$. Second, the MSE-trained GNNHAR rows do not improve on GHAR$_M$. Third, QLIKE-trained models are uniformly stronger under QLIKE forecast loss. GHAR$_Q$, GNNHAR2L$_Q$, and GNNHAR3L$_Q$ all achieve QLIKE ratios around 0.976 to 0.979.

Model	Block	EC	ρ_M	ρ_Q	$1-\rho_Q$
GHAR_M	strict core	M	0.9975	0.9925	0.75%
GNNHAR1L_M	strict core	M	1.0000	0.9968	0.32%
GNNHAR2L_M	strict core	M	1.0001	0.9995	0.05%
GNNHAR3L_M	strict core	M	1.0011	0.9996	0.04%
HAR_M	strict core	M	1.0000	1.0000	0.00%
GHAR_Q	strict core	Q	1.0028	0.9764	2.36%
GNNHAR1L_Q	strict core	Q	1.0039	0.9802	1.98%
GNNHAR2L_Q	strict core	Q	1.0035	0.9795	2.05%
GNNHAR3L_Q	strict core	Q	1.0030	0.9774	2.26%
HAR_Q	strict core	Q	1.0056	0.9801	1.99%
GHAR_M^{IV}	IV extension	M	0.9649	0.9778	2.22%
GNNHAR1L_M^{IV}	IV extension	M	0.9716	0.9868	1.32%
GNNHAR2L_M^{IV}	IV extension	M	0.9888	1.0122	-1.22%
GNNHAR3L_M^{IV}	IV extension	M	0.9676	0.9900	1.00%
HAR_M^{IV}	IV extension	M	0.9648	0.9792	2.08%
GHAR_Q^{IV}	IV extension	Q	0.9812	0.9335	6.65%
GNNHAR1L_Q^{IV}	IV extension	Q	0.9814	0.9349	6.51%
GNNHAR2L_Q^{IV}	IV extension	Q	0.9823	0.9335	6.65%
GNNHAR3L_Q^{IV}	IV extension	Q	0.9815	0.9401	5.99%
HAR_Q^{IV}	IV extension	Q	0.9807	0.9328	6.72%
GNNHAR4L_M	depth extension	M	1.0040	1.0032	-0.32%
GNNHAR4L_M^{IV}	depth extension	M	0.9789	0.9955	0.45%
GNNHAR5L_M	depth extension	M	1.0030	1.0105	-1.05%
GNNHAR5L_M^{IV}	depth extension	M	0.9796	0.9936	0.64%
GNNHAR4L_Q	depth extension	Q	1.0026	0.9785	2.15%
GNNHAR4L_Q^{IV}	depth extension	Q	0.9858	0.9361	6.39%
GNNHAR5L_Q	depth extension	Q	1.0033	0.9768	2.32%
GNNHAR5L_Q^{IV}	depth extension	Q	0.9809	0.9342	6.58%

Table 7: Current SP100 A100 full available model comparison. Ratios are relative to HAR_M.

Model	EC	ρ_M	ρ_Q	$1-\rho_Q$
HAR_M	M	1.0000	1.0000	0.00%
GHAR_M	M	0.9975	0.9925	0.75%
GNNHAR1L_M	M	1.0000	0.9968	0.32%
GNNHAR2L_M	M	1.0001	0.9995	0.05%
GNNHAR3L_M	M	1.0011	0.9996	0.04%
HAR_Q	Q	1.0056	0.9801	1.99%
GHAR_Q	Q	1.0028	0.9764	2.36%
GNNHAR1L_Q	Q	1.0039	0.9802	1.98%
GNNHAR2L_Q	Q	1.0035	0.9795	2.05%
GNNHAR3L_Q	Q	1.0030	0.9774	2.26%

Table 8: Current SP100 A100 strict Zhang-core loss ratios relative to HAR_M.

This pattern is close to Zhang et al. on the QLIKE point but not on the nonlinear-spillover point. The current SP100 result supports QLIKE training, while it does not give a clean dominance of GNNHAR1L over GHAR. The best strict-core QLIKE row is GHAR_Q or a deeper QLIKE GNN row by a very small margin depending on the exact loss considered. Those differences are economically small relative to the gap between MSE-trained and QLIKE-trained specifications.

Table 9 gives the IV-augmented extension. The IV extension is substantially stronger than the strict core. HAR_Q^{IV}, GHAR_Q^{IV}, and GNNHAR2L_Q^{IV} all reduce QLIKE loss by roughly 6.6% to 6.7% relative to HAR_M. This is the most stable empirical gain in the current long run. It should, however, be described as an extension to Zhang's setup rather than a direct replication of the paper.

Model	EC	ρ_M	ρ_Q	$1-\rho_Q$
HAR_M^{IV}	M	0.9648	0.9792	2.08%
GHAR_M^{IV}	M	0.9649	0.9778	2.22%
GNNHAR1L_M^{IV}	M	0.9716	0.9868	1.32%
GNNHAR2L_M^{IV}	M	0.9888	1.0122	-1.22%
GNNHAR3L_M^{IV}	M	0.9676	0.9900	1.00%
HAR_Q^{IV}	Q	0.9807	0.9328	6.72%
GHAR_Q^{IV}	Q	0.9812	0.9335	6.65%
GNNHAR1L_Q^{IV}	Q	0.9814	0.9349	6.51%
GNNHAR2L_Q^{IV}	Q	0.9823	0.9335	6.65%
GNNHAR3L_Q^{IV}	Q	0.9815	0.9401	5.99%

Table 9: Current SP100 A100 IV-augmented extension loss ratios relative to HAR_M.

5.3 SP500 Scale Evidence

The SP500 artifact in Table 10 is not a completed A100 long run. It is included because it answers whether the existing pipeline can be pushed to a much larger graph. Table 10 gives the full available SP500 scale comparison, including diagnostics. Table 11 gives the strict core. Under the short-budget scale protocol, GHAR is nearly tied with HAR, while GNNHAR1L and deeper GNNHAR models are worse under QLIKE. This artifact therefore does not support the claim that increasing the number of nodes strengthens the GNNHAR gain.

Model	Block	Est.	ρ_M	ρ_Q	$1-\rho_Q$	ρ_Q^{IV}
GHAR	strict core	MSE	0.9985	0.9993	0.07%	1.0271
HAR	strict core	MSE	1.0000	1.0000	0.00%	1.0278
GNNHAR1L	strict core	MSE	1.0085	1.0331	-3.31%	1.0618
GNNHAR2L	strict core	MSE	1.1022	1.1374	-13.74%	1.1690
GNNHAR3L	strict core	MSE	1.2913	1.3972	-39.72%	1.4361
GHAR+IV	IV extension	MSE	0.9635	0.9727	2.73%	0.9998
HAR+IV	IV extension	MSE	0.9639	0.9729	2.71%	1.0000
GNNHAR3L-IV	IV extension	MSE	1.0820	1.1144	-11.44%	1.1454
GNNHAR1L-IV	IV extension	MSE	1.0654	1.1147	-11.47%	1.1457
GNNHAR2L-IV	IV extension	MSE	1.1760	1.2686	-26.86%	1.3039
GHAR+fakeIV	diagnostic placebo	MSE	0.9904	0.9947	0.53%	1.0223
HAR+fakeIV	diagnostic placebo	MSE	0.9935	0.9960	0.40%	1.0237
GNNHAR1L-IV+fakeIV	diagnostic placebo	MSE	1.3553	1.4775	-47.75%	1.5186
GHAR+IV-random	diagnostic random graph	MSE	0.9638	0.9733	2.67%	1.0003
GHAR_RANDOM	diagnostic random graph	MSE	0.9982	0.9978	0.22%	1.0255
GNNHAR3L-IV-random	diagnostic random graph	MSE	1.2192	1.3049	-30.49%	1.3412

Table 10: SP500 scale-run full available model comparison. Diagnostic placebo and random-graph rows are included for completeness.

Model	ρ_M	ρ_Q	$1-\rho_Q$
HAR	1.0000	1.0000	0.00%
GHAR	0.9985	0.9993	0.07%
GNNHAR1L	1.0085	1.0331	-3.31%
GNNHAR2L	1.1022	1.1374	-13.74%
GNNHAR3L	1.2913	1.3972	-39.72%

Table 11: SP500 scale-run strict-core loss ratios relative to HAR. This is a short-budget scale artifact, not the latest A100 protocol.

The IV extension in Table 12 again improves the linear benchmarks but does not rescue the GNN rows in this short-budget run. HAR^{IV} and GHAR^{IV} reduce QLIKE loss by about 2.7%, while IV-augmented GNNHAR rows are worse than HAR. This result should be treated cautiously until an SP500 A100 long run is completed.

Model	ρ_M	ρ_Q	$1-\rho_Q$
HAR^{IV}	0.9639	0.9729	2.71%
GHAR^{IV}	0.9635	0.9727	2.73%
$\mathrm{GNNHAR1L}^{IV}$	1.0654	1.1147	-11.47%
$\mathrm{GNNHAR2L}^{IV}$	1.1760	1.2686	-26.86%
$\mathrm{GNNHAR3L}^{IV}$	1.0820	1.1144	-11.44%

Table 12: SP500 scale-run IV extension loss ratios relative to HAR.

6 Forecast Evaluation and DM Tests

Zhang et al. use MCS to identify a set of statistically competitive models and DM tests to compare forecast losses, with special emphasis on QLIKE because it has stronger power for volatility forecast comparison. Table 13 reports the MCS summaries available in our completed artifacts after excluding diagnostic fake-IV and random-graph rows. The Dow30 MCS set remains broad, including IV, HAR, GHAR, and some GNN variants. The SP100 and SP500 scale MCS sets are narrow and dominated by HAR^{IV} and GHAR^{IV}.

Universe	Models retained in 5% MCS
Dow30	GHAR+IV, GNNHAR3L-IV, HAR+IV, HAR, GHAR, GNNHAR2L-IV ...
SP100 scale	GHAR+IV, HAR+IV
SP500 scale	GHAR+IV, HAR+IV

Table 13: Model Confidence Set summaries after excluding diagnostic random-graph and fake-IV rows.

Table 14 gives matched QLIKE DM tests for the main questions. The sign convention is that B is the second model in the comparison, and $1 - L_B/L_A > 0$ favors B. The graph-linearity comparison HAR $\rightarrow$ GHAR is positive but not statistically significant in the completed Dow30, SP100-scale, and SP500-scale artifacts. The nonlinear comparison GHAR $\rightarrow$ GNNHAR1L is not significant on Dow30 and is significantly negative in the scale artifacts. The IV-matched comparison $\mathrm{GHAR}^{IV} \rightarrow \mathrm{GNNHAR1L}^{IV}$ is significantly negative in all three available artifacts.

Universe	Question	Comparison	L_B/L_A	$1 - L_B/L_A$	DM	p	Result
Dow30	graph linearity	HAR vs GHAR	0.9868	1.32%	0.4670	0.6411	not significant
Dow30	nonlinearity	GHAR vs GNNHAR1L	1.0442	-4.42%	-1.5033	0.1345	not significant
Dow30	nonlinearity with IV	GHAR+IV vs GNNHAR1L-IV	1.0634	-6.34%	-2.1735	0.0310	deteriorates
SP100 scale	graph linearity	HAR vs GHAR	0.9977	0.23%	1.2934	0.1977	not significant
SP100 scale	nonlinearity	GHAR vs GNNHAR1L	1.0184	-1.84%	-4.4868	$< 10^{-4}$	deteriorates
SP100 scale	nonlinearity with IV	GHAR+IV vs GNNHAR1L-IV	1.1120	-11.20%	-13.4857	0.0000	deteriorates
SP500 scale	graph linearity	HAR vs GHAR	0.9993	0.07%	0.5480	0.5844	not significant
SP500 scale	nonlinearity	GHAR vs GNNHAR1L	1.0339	-3.39%	-8.9413	$< 10^{-4}$	deteriorates
SP500 scale	nonlinearity with IV	GHAR+IV vs GNNHAR1L-IV	1.1459	-14.59%	-19.7360	0.0000	deteriorates

Table 14: Matched QLIKE DM tests in available completed artifacts. B is the second model in the comparison; positive $1 - L_B/L_A$ favors B.

This is where our conclusion necessarily diverges from Zhang et al. Their paper reports that nonlinear GNNHAR improves on linear GHAR, especially for short horizons, and that QLIKE training improves performance. Our current evidence supports the QLIKE point and supports IV as a valuable extension, but it does not support the nonlinear GNN gain over GHAR under the VolVue 30-day target.

7 Discussion: Large Graphs, Depth, and Multi-Hop Information

The natural large-graph question is not whether the SP100 or SP500 raw loss level is lower than the Dow30 loss level. Those losses are computed on different assets and different volatility distributions. The better question is whether the marginal gain from nonlinear graph aggregation increases when the graph has more nodes:

$$\Delta_{\mathrm{GNN}} = 1 - \frac{L(\mathrm{GNNHAR})}{L(\mathrm{GHAR})}.$$

The matched DM tests do not support such a claim at present. The larger scale artifacts show a more negative GNNHAR1L-vs-GHAR comparison, not a stronger positive one. Therefore, in the current report we should not argue that larger N improves the GNN contribution. At most, we can say that the question remains open pending a matched SP500 A100 long run.

Table 15 reports the current SP100 A100 depth comparisons. The Zhang-style core comparison $\mathrm{GNNHAR1L}_Q \rightarrow \mathrm{GNNHAR2L}_Q$ is slightly positive but not statistically significant. The MSE-trained core comparison is slightly negative and not significant. In the IV extension, the direction varies by layer and criterion. Thus, we do not find a monotone depth improvement.

Base	Candidate	IV	EC	L_B/L_A	$1 - L_B/L_A$	DM	p	Result
$\mathrm{GNNHAR1L}_M$	$\mathrm{GNNHAR2L}_M$	No	M	1.0027	-0.27%	-1.2066	0.2276	not significant
$\mathrm{GNNHAR1L}_Q$	$\mathrm{GNNHAR2L}_Q$	No	Q	0.9992	0.08%	0.6368	0.5242	not significant
$\mathrm{GNNHAR1L}_M^{IV}$	$\mathrm{GNNHAR2L}_M^{IV}$	Yes	M	1.0257	-2.57%	-2.6328	0.0085	deteriorates
$\mathrm{GNNHAR2L}_M^{IV}$	$\mathrm{GNNHAR3L}_M^{IV}$	Yes	M	0.9781	2.19%	2.5408	0.0111	improves
$\mathrm{GNNHAR2L}_Q^{IV}$	$\mathrm{GNNHAR3L}_Q^{IV}$	Yes	Q	1.0071	-0.71%	-2.0867	0.0369	deteriorates

Table 15: Current SP100 A100 GNN depth DM tests. Positive $1 - L_B/L_A$ favors the candidate model.

Zhang et al. explicitly warn that too many GNN layers may lead to over-smoothing. Our current SP100 result is consistent with that caution: deeper GNNs are not uniformly better, and any five-layer improvement should be treated as an extension result rather than the core replication result.

8 Robustness and Extensions

The additional models in Table 16 and Table 17 are useful for robustness, but they should not be mixed into the main Zhang-core table. The four- and five-layer models follow Zhang et al.'s larger-universe logic for S&P 100, where the graph diameter may require more layers to cover all nodes. The linear multi-hop GHAR variants correspond to the same question as Zhang et al.'s Appendix E: do explicit two-hop linear neighbors add predictive power beyond one-hop GHAR?

Model	EC	ρ_M	ρ_Q	$1 - \rho_Q$
$\mathrm{GNNHAR4L}_M$	M	1.0040	1.0032	-0.32%
$\mathrm{GNNHAR5L}_M$	M	1.0030	1.0105	-1.05%
$\mathrm{GNNHAR4L}_Q$	Q	1.0026	0.9785	2.15%
$\mathrm{GNNHAR5L}_Q$	Q	1.0033	0.9768	2.32%
$\mathrm{GNNHAR4L}_M^{IV}$	M	0.9789	0.9955	0.45%
$\mathrm{GNNHAR5L}_M^{IV}$	M	0.9796	0.9936	0.64%
$\mathrm{GNNHAR4L}_Q^{IV}$	Q	0.9858	0.9361	6.39%
$\mathrm{GNNHAR5L}_Q^{IV}$	Q	0.9809	0.9342	6.58%

Table 16: Current SP100 A100 four- and five-layer depth extensions. These are outside Zhang's main DJIA core but follow his larger-universe extension logic.

Base	Candidate	IV	EC	L_B/L_A	$1 - L_B/L_A$	DM	p	Result
linear GHAR 1-hop	linear GHAR 1+2-hop	No	M	1.0055	-0.55%	-4.0647	$< 10^{-4}$	deteriorates
linear GHAR 1-hop	linear GHAR 1+2+3-hop	No	M	1.0082	-0.82%	-4.0762	$< 10^{-4}$	deteriorates
GNNHAR1L$_Q$	GNNHAR5L$_Q$	No	Q	0.9965	0.35%	2.2236	0.0262	improves
GNNHAR4L$_Q$	GNNHAR5L$_Q$	No	Q	0.9983	0.17%	2.0347	0.0419	improves

Table 17: SP100 A100 robustness DM tests for linear multi-hop GHAR and four-/five-layer GNN extensions.

In our SP100 A100 run, the linear multi-hop GHAR extension is significantly worse than one-hop GHAR. The two-hop linear candidate has $L_B/L_A = 1.0055$, and the three-hop linear candidate has $L_B/L_A = 1.0082$. Both are statistically negative at conventional levels. This is directionally consistent with Zhang et al.'s conclusion that multi-hop neighbors do not provide clear additional predictive power once zero-hop and one-hop information are included. The stronger difference is that our multi-hop extension is significantly worse, whereas Zhang et al.'s appendix-style GHAR2Hop DM result is not statistically significant.

9 Conclusion

The Zhang-style reading of our current results is precise. First, the pipeline now implements the intended HAR–GHAR–GNNHAR comparison and records loss ratios, MCS summaries, and DM tests in a form that can be reported. Second, the current VolVue target is not Zhang's high-frequency realized variance, so weaker GNN results are not automatically evidence of a code failure. The 30-day historical-volatility proxy is smoother and already embeds lagged information.

Third, QLIKE training is the part of Zhang et al.'s empirical message that transfers most clearly to our current SP100 long run. QLIKE-trained strict-core models have much lower QLIKE ratios than their MSE-trained counterparts. Fourth, implied volatility is the strongest extension in our data: IV-augmented HAR/GHAR/GNNHAR rows dominate many strict-core rows, especially in the SP100 A100 run.

Fifth, the nonlinear GNNHAR gain over GHAR is not supported by the current evidence. The completed DM tests are not favorable to GNNHAR1L relative to GHAR, and the IV-matched GNN comparisons are significantly negative. We therefore should not write the paper as if our results replicate Zhang et al.'s nonlinear-spillover finding. A more accurate framing is that we conduct a Zhang-style extension to VolVue 30-day HV/IV proxies and find that QLIKE and IV information matter more robustly than nonlinear graph depth under the current target.

For a publication-quality next pass, the most important improvement is data construction. Either build a daily close-to-close squared-return target

$$RV_{i,t}^{cc} = \left(100 \log \frac{P_{i,t}}{P_{i,t-1}}\right)^2$$

from adjusted closes, or obtain high-frequency intraday data to construct true realized variance. Then rerun the same Zhang-style rolling protocol for Dow30, SP100, and SP500. If we keep the VolVue target, the manuscript should explicitly call the study an extension to 30-day volatility proxies, not a direct realized-volatility replication.

A Artifacts

The report source is

`reports/gnnhar_results_20260615/gnnhar_results_report.tex`.

Generated report tables are under

`reports/gnnhar_results_20260615/tables/`.

The local notebook backup is

`notebooks/snapshots/GNNHAR_Colab_Full_Run_local_backup_20260615.ipynb`.

The Zhang paper was read through MinerU Markdown extraction at

`references/papers/zhang-chao-recent/mineru-verify/gnnhar-ijf-2025/`.